

Network Breakdown

Time limit: 1.00 s

Memory limit: 512 MB

Syrjälä's network has n computers and m connections between them. The network consists of components of computers that can send messages to each other.

Nobody in Syrjälä understands how the network works. For this reason, if a connection breaks down, nobody will repair it. In this situation a component may be divided into two components.

Your task is to calculate the number of components after each connection breakdown.

Input

The first input line has three integers n , m and k : the number of computers, connections and breakdowns. The computers are numbered 1, 2, ..., n .

Then, there are m lines describing the connections. Each line has two integers a and b : there is a connection between computers a and b . Each connection is between two different computers, and there is at most one connection between two computers.

Finally, there are k lines describing the breakdowns. Each line has two integers a and b : the connection between computers a and b breaks down.

Output

After each breakdown, print the number of components.

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq k \leq m$
- $1 \leq a, b \leq n$

Example

Input	Output
5 5 3 1 2 1 3 2 3 3 4 4 5 3 4 2 3 4 5	2 2 3